THE SPECIALIZATION AND COSPECIALIZATION MAPS FIT INTO A COMMUTATIVE DIAGRAM OF COHOMOLOGY GROUPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to the strict henselisation, the simplification of the three maps, the identification of the composite through beta, and the conclusion that cosp inverts the same identification.

Editorial note. The label lemma-specialization-cospecialization names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's commutative diagram is the author's, delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- The comparison map beta and its identification with the stalk-level alpha. Same chapter, the lemma identifying the pullback of beta with the map alpha used in the definition of local acyclicity. Supplies the description of beta used to compute the composite in the diagram.
- Higher direct images as cohomology of the fibre. Same chapter, the theorem computing the stalk of the pushforward as the cohomology of the geometric fibre. Supplies the identification (Rf_*K)_s = R Gamma(X, K) after the base change to the strict henselisation.

Source reference. Original reference 0GJU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19420–19420. Prerequisite pointer invoked at proof line(s) 19484; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03Q9: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8015–8015. Prerequisite pointer invoked at proof

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line(s) 19490; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above, if in addition f is quasi-compact and quasi-separated, then the diagram*

$$\begin{array}{ccc} (Rf_*K)_{\overline{s}} & \longrightarrow & R\Gamma(X_{\overline{s}}, K) \\ {\scriptstyle sp}\downarrow & & \uparrow{\scriptstyle cosp} \\ (Rf_*K)_{\overline{t}} & \longrightarrow & R\Gamma(X_{\overline{t}}, K) \end{array}$$

is commutative.

Proof. As in the proof of Lemma 0GJU we may replace S by $\mathrm{Spec}(\mathcal{O}^{sh}_{S,\overline{s}})$. Then our maps simplify to $h : X_{\overline{t}} \to X$, $i : X_{\overline{s}} \to X$, and $\beta_{K,\overline{s},\overline{t}} : K \to Rh_*(K|_{X_{\overline{t}}})$. Using that $(Rf_*K)_{\overline{s}} = R\Gamma(X, K)$ by Theorem 03Q9 the composition of sp with the base change map $(Rf_*K)_{\overline{t}} \to R\Gamma(X_{\overline{t}}, K)$ is just pullback of cohomology along h. This is the same as the map

$$R\Gamma(X, K) \xrightarrow{\beta_{K,\overline{s},\overline{t}}} R\Gamma(X, Rh_*(K|_{X_{\overline{t}}})) = R\Gamma(X_{\overline{t}}, K)$$

Now the map $cosp$ first inverts the $=$ sign in this displayed formula, then pulls back along i, and finally applies the inverse of $i^{-1}\beta_{K,\overline{s},\overline{t}}$. Hence we get the desired commutativity. □